(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: ECC-207 **Subject: Digital Logic & Computer Design**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data, if any.

Q1 Attempt all questions: **(3×5=15)**

a) Reduce the Boolean Expression $+^{-}(+^{-})$

b) What are don't care terms?

c) Differentiate a decoder from a Demultiplexer.

d) Draw the state table and excitation table of T flip flop.

e) Write about Virtual memory.

UNIT-I

Q2 a) What is meant by duality in Boolean algebra? State and prove Associative and Distributive theorems. **(7)**

b) Minimize the expression $(\ ,\ ,\ ,\ ) = \sum(0,1,2,5,8,9,10)$ using K-map and obtain (i) minimal SOP and (ii) minimal POS expression. **(8)**

Q3 a) Explain the operation of a magnitude comparator? **(7)**

b) Implement a full subtractor using 8:1 MUX. **(8)**

UNIT-II

Q4 a) What is Race Around condition? And how is it overcome in JK Flip-Flop? Explain with diagram. **(5)**

b) Design a Mod-5 synchronous counter using JK Flip-Flop. **(10)**

Q5 a) Explain designing of T Flip-Flop using JK Flip-Flop. **(7)**

b) Write short notes on: (i) Shift registers (ii) PLA & PAL **(8)**

UNIT-III

Q6 a) Differentiate between hardwired control and micro programmed control and explain organizations of micro programmed control unit with neat diagram. **(10)**

b) What is addressing mode? Explain different types of addressing modes. **(5)**

Q7 a) What are the different phases of an instruction cycle? Explain with the help of flowchart. **(7)**

b) What is assembly language? Explain basic computer instruction formats with suitable diagram. **(8)**

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UNIT-IV

Q8 a) Define DMA. Explain its need. Explain DMA transfer in detail with the help of suitable diagram. **(7)**

b) What is an interrupt? Explain interrupt cycle with the help of flowchart. **(8)**

Q9 Write short notes on:

a) Modes of data transfer **(5)**

b) Cache memory **(5)**

c) Associative memory **(5)**

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(Please write your Enrolment No. immediately) Enrolment No.....

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

3RD Semester, November 2022

Paper Code:ECC-207 Subject: Digital Logic and Computer Design

Time: 1.5 hrs. Max Marks: 30

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Question 1

a. How is BCD Code different from Binary Code? (2 Marks) CO1
b. Explain addition of two negative numbers in 2's compliment form with help of an example. (2 Marks) CO1
c. Implement two input EX-OR gate using a Multiplexer? (2 Marks) CO1
d. Explain the concept of Level Triggering in Sequential Circuits? (2 Marks) CO2
e. How is state diagram different from state table? (2 Marks) CO 2

Question 2

a. Differentiate between Serial Adder and Binary Parallel Adder? (3 Marks), CO 1,2
b. Explain how to Convert SR Flip Flop into D Flip Flop. (3 Marks), CO 2
c. Design a 4 bit Ripple Down Counter and draw the waveform? (4 Marks), CO 2

Question 3

a. For the Boolean expression $F = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + A\bar{B}\bar{C} + ABC + AB\bar{C}$ find out the minimized Product of Sum (POS) expression (use K-map) (5 Marks), CO1
b. What is a Multiplexer? Design a 16:1 multiplexer using 4:1 multiplexers? (5 Marks), CO1

c. **Question 4**

a. Using K-map simplify expression, Y= f(A,B,C,D) = $\pi(0,1,2,4,5,8,9,15)$ (2 Marks), CO 1
b. A combinational circuit is defined by function $F_1 = \sum_m (1, 5, 7)$, $F_2 = \sum_m (5, 6, 7)$. Implement the circuit with a PLA. (3 Marks), CO 2
c. Design a Mod-5 asynchronous counter using T-FF. (5 Marks), CO 2

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

3rd Semester, November, 2022

Paper Code: ECC-209 **Subject: Analog Communications**

Time: 1½ Hrs. **Max. Marks: 30**

Note: Attempt Q. No. 1 which is compulsory and any two more questions from remaining.

Q. No.	Question	Max. Marks	CO(s)
1 (a)	Derive an expression for a single tone frequency modulation.	2	1, 2
1 (b)	The signal is modulated at 7.5 V and carrier voltage is of 9V, calculate modulation percentage.	2	1, 2
1 (c)	A carrier signal is frequency modulated with a sinusoidal signal of 2 KHz resulting in a maximum frequency deviation of 5 KHz. Find the bandwidth of the modulated signal.	2	1, 2
1(d)	In the demodulation process of DSB-FC using envelope detector, how the value of time constant RC determmine the distortion in received signal.	2	1, 2
1(e)	Compare the SSB-SC and DSB-SC in termis of bandwidth and power saving. Also give the practical application of VSB.	2	1, 2
			1, 2
2 (a)	Explain the Phase Locked Loop Frequency demodulator with derivation.	5	1, 2
2(b)	Differentiate between NBFM and WBFM. Where they are used? What are the merits and Limitation of FM.	5	1, 2
3 (a)	An AM transmitter radiates 9kW power when the carrier is unmodulated and 10.125kW when carrier is sinusoidally modulated. Find the modulation index. Now if another sine wave coresponding to 40 % modulation is transmitted simultaneously, calculate the total radiated power.	5	1, 2
3 (b)	Explain the suppression of carrier in generation of DSB-SC with suitable diagram using balanced Modulator.	5	1, 2
4 (a)	Explain Superhetrodyne receiver with block diagram.	5	1, 3
4 (b)	b) A carrier voltage has a peak amplitude of 10 V at a frequency of 1 MHz. A sinusoidal signal of 1 kHz varies the amplitude of RF wave between 7.5 V and 12.5 V. Find the modulation index and equation of AM wave.	5	1, 2